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Unit II - Process & Thread Management

process:

A process is basically a program while it is being executed.

- A process is more than program code.
- It also include the following.
- Value of program counter
- Contents of CPU register.
- program code
- Data Section containing global variables.
- Dynamically allocated memory during runtime.
- A program in itself is not process.
- A process is basically a program execution.
- A process is defined as an entity which represents the basic unit of work to be implemented in the system.

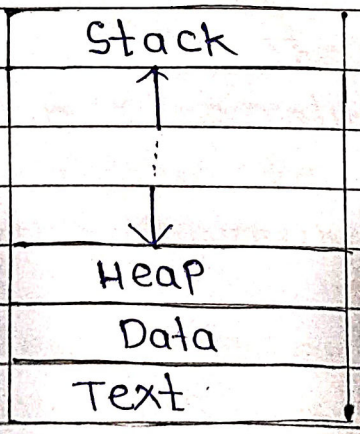

Fig. process in memory.

- Stack - The process stack contains the temporary data such as method/function parameters, local variables
- Heap - This is dynamically allocated memory to a process during its run time.
- Data - This section contains the global & static variables.

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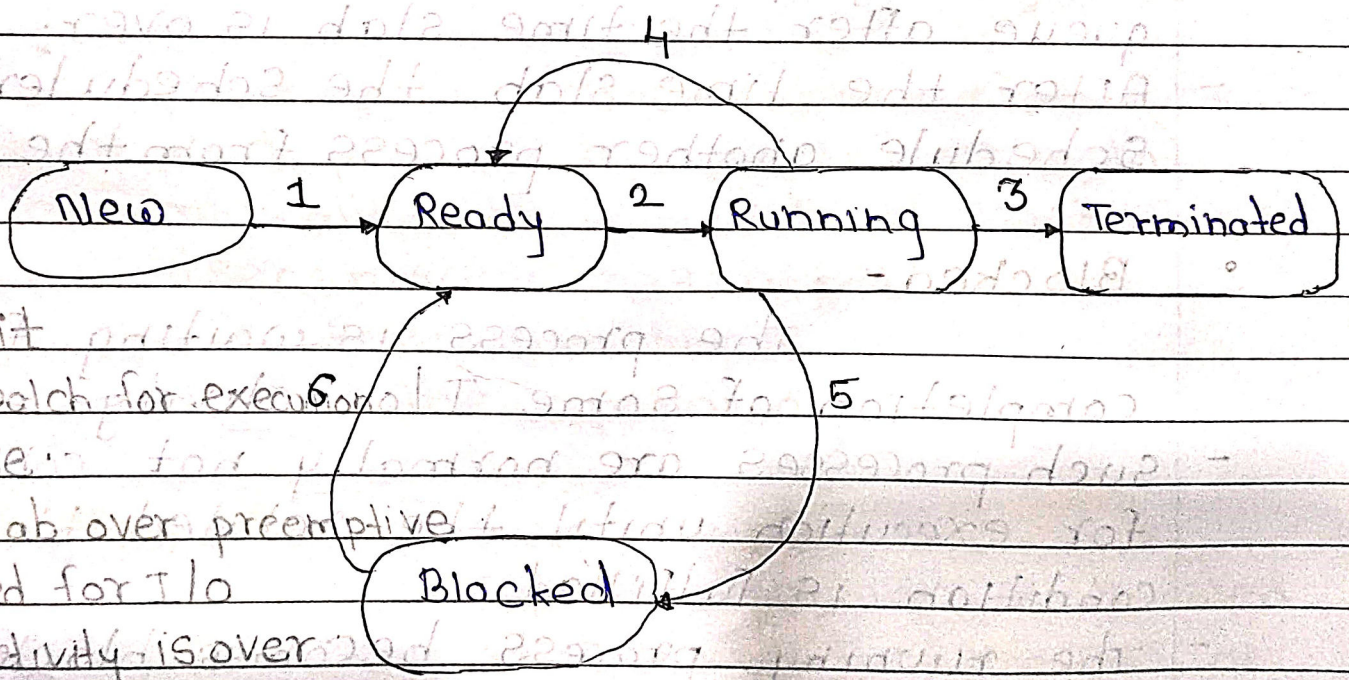
process States and Transitions diagram

The lifetime of a process can be divided into several stages as states, each with certain characteristics that describe the process.

- Each process may be in one of the following states.

- New
- Ready
- Running
- Blocked
- Terminated



1) Admit
2) Dispatch for execution
3) Release
4) Time slab over preemptive
5) Blocked for I/o
6) I/o activity is over

Fig. process State transition diagram

- New - The process has been created.

- Ready - The process is waiting to be allocated to a processor.

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- processor comes to this state immediately after it is created.
- All ready processes are kept in queue.
- A program called scheduler, which is a part of the operating system, picks up one process for execution by passing control it.

- Running-

A CPU is currently allocated to the process & the process getting executed

- A running process is blocked if there is an I/o request during execution.
- A running process is sent to the ready queue after the time slab is over.
- After the time slab, the scheduler may schedule another process from the ready

- Blocked-

The process is waiting for completion of some I/o activity.

- Such processes are normally not considered for execution until the related blocking condition is fulfilled.
- The running process becomes blocked by invoking I/O routines, whose result is needed during execution.

- Terminated-

The process has finished its execution-

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- Terminated -
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Process Control Block (PCB)

Process ID
State
Pointer
Priority
Program Counter
CPU registers
I/O Information
Accounting Information
etc.....

Fig. Structure of process control Block

- The operating system maintains the information about each process in a record or a data structure called process control Block (PCB).
- Each user process has a PCB.
- All these PCBs are kept in the memory reserved for the operating system.

- **Process ID**-

process ID is a number allocated by the operating system to the process on creation.

- operating system normally set a limit on the maximum number of processes that it can handle & schedule.
- A unique identifier for each process.

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- **Process state** - The state may be new, ready, running, waiting, halted & so on.

- **process priority** - The process which are urgently required to be completed are set at higher priority while are set with others (lower priority).
  - This priority can be set externally by the user/system manager, or it can be decided by the operating system internally depending on various parameters.

- **program counter** - It gives the address of the next instruction to be executed for this process.

- **Register Save Area** - This various CPU registers where process need to be stored for execution for running state.

- **Accounting Information** - This include the amount of CPU used for process execution, time limits, execution ID etc.

- **I/O status Information** - Information about I/O devices allocated to the process.

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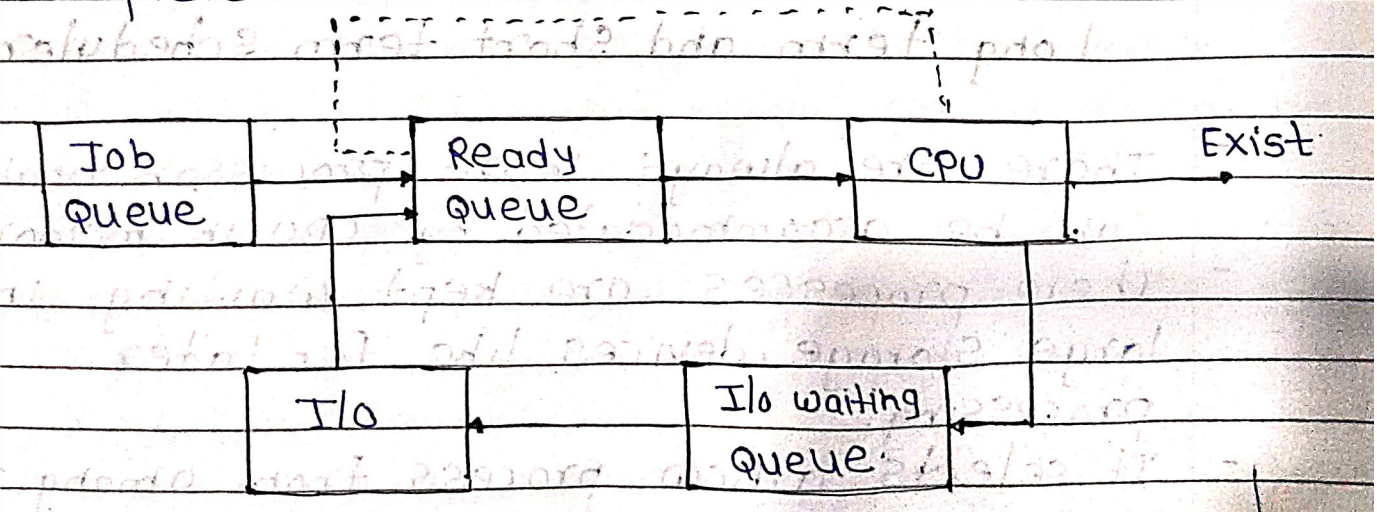
- **Memory Management Information**- Details about the memory allocated to the process, such as page tables, or segment tables.
- **CPU Register**- Contains the value of the CPU registers used by the process.
- **Open Files**- Information about files currently open by the process.
- **Pointer**- This gives pointers to other data structures maintained for that process.
- **Process Scheduling**:-

process scheduling is a fundamental operating system function.

- Scheduling is a set of policies & mechanisms supported by operating system that controls the order in which the work to be done is completed.



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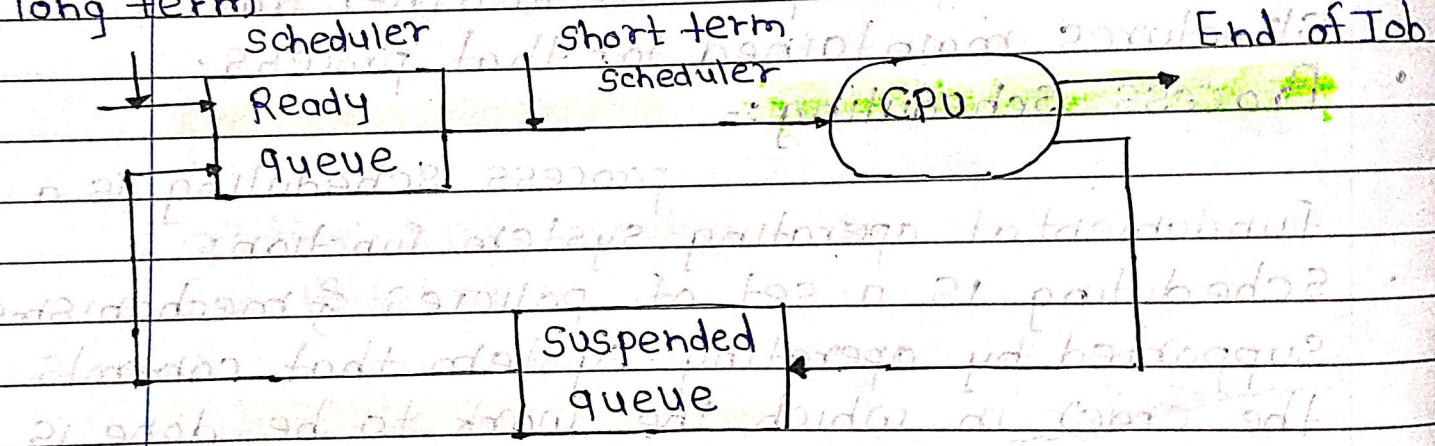
Types of Schedulers

- There are three types of schedulers.
1) Long term schedulers
2) Medium term schedulers
3) short term schedulers

- **Long term scheduler**
- Long term Scheduler is also called Job Scheduler.
- The determines which job shall be admitted for processing.



Long term and Short term scheduler.

- There are always more processor than it can be accommodated by CPU in memory.
- There processes are kept waiting in large storage devices like for later processing.
- It selects a new process from among the waiting processes using SJF or FCFS.

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Strategy:
- If all processes are CPU bound then system response will become poor.
- If all processes are I/O bound, the ready queue will always be empty.

- **Short term Scheduler**-

It allocates processes belonging to ready queue to CPU for immediate processing.
- Short term scheduler is called very often:
- Called at the end of time slab.
- Called in case of I/O Request

- **Medium term Scheduler**:-

Most of the processes require I/o operation.
- In that case, it may become Suspended for I/o operation after running a while.
- It is beneficial to remove these processes from main memory to hard hard disk to create more memory for other proeress.
- At some later time these processes can be reloaded into memory.
- The suspended processes are swapped out & swapped in by medium term scheduler.

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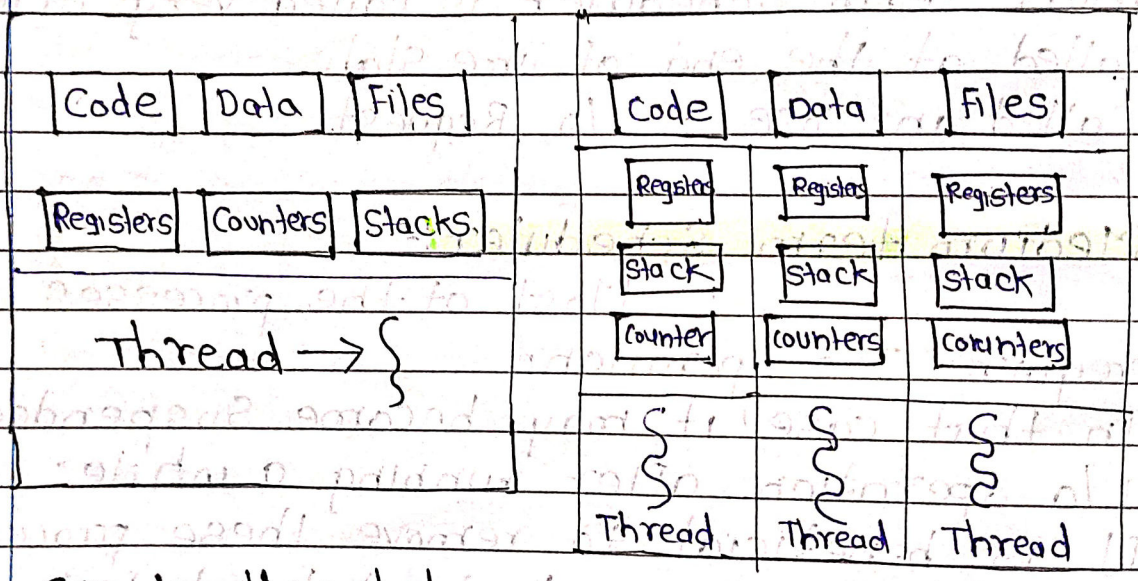
Threads

Thread is a sequential flow of tasks within a process.
- Thread is an operating system can be of the same or different types.
- Threads are used to increase the performance of the applications.
- Each thread has its own program counter, stack & set of registers.



Single-threaded process | Multithread process

- A thread has the following three components
- program counter
- Register set
- Stack space.
- In multithreading, the idea is to divided a single process into multiple threads instead of creating a whole new process.

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- The other advantages of multithreading are mentioned below.
- Resource sharing - Thread of a single process share the same resource such as code, data / file.
- Responsiveness
- Economy
- A multi-threaded process contain multiple threads of execution with a single process.

Single-Thread processes:

- A single-threaded process contains only one thread of execution.
- This means it executes instructions sequentially, one
- It has a single program counter, a single stack & a single set of CPU registers.
- Task are process linearly & one task must complete before the next can begin.

Multithreading process:

In multithreading, the idea to divide a single process into multiple threads instead of creating a new process.

- Multithreading is an operating system allow multiple part of a program called threads, to execute concurrently within the same process.
- This enable improve performance responsiveness, & resource utilization on system with multiple CPU cores.

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	User level Thread	kernel level Thread
1)	User thread are implemented by user	kernel level thread are implemented by OS.
2)	Context switch time is less.	context switch time is more.
3)	Context switch require no hardware support	Hardware support is required.
4)	Implementation of user thread is easy.	Implementation of user thread is complicated.
5)	user level threads are faster to create & manage.	kernel level threads are slower to create & manage.
6)	User level threads are also called many to one mapping threads	kernel level thread support one to many thread mapping
7)	user level thread can ran on any operating System.	kernel level thread are specific to the operating System.
8)	Example - Java Thread	Example - Windows 95/98/NT Digital UNIX.

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Differentiate between process & Thread

	process	Thread
1)	process mean any program is in execution.	Threads means a segment of a process.
2)	The process takens more time to terminates.	The thread takes less time to terminate.
3)	It takes more time for creation.	It takes less time for creation.
4)	The process is isolated.	Thread share memory
5)	The process does not share data with each other.	Threads share data with each other.
6)	The process is less efficient in terms of communication.	Threads is more efficient in terms of communication.
7)	It also takes more time for context switching.	It takes less time for context switching.
8)	A system call is Involved in It.	No system call is involved.

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Scheduling Algorithm -

- First come First serve (FCFS)
- Shortest Job First (SJF) scheduling
- priority scheduling
- Round Robin (RR) scheduling

First come First serve (FCFS)

- Jobs are executed on first come, first serve basis.
- It is a non-preemptive, pre-emptive scheduling algorithm.
- Easy to understand & implement.
- poor in performance as average wait time is high.
- FCFS stand for First come First serve.
- Convey effect is a phenone associated with the FCFS algorithm, in which whole os slow down due to Few process.

- Example

process	Burst Time
P1	24
P2	3
P3	4

Calculate average waiting time.
Calculate Average Turn around Time.

→ Gantt chart → is a horizontal bar chart to demostrate CPU scheduling.

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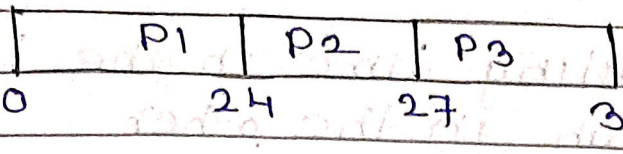

- Waiting Time -

P1 = 0

P2 = 24 $= \frac{0+24+27}{3} = 17$

P3 = 27

- Turn around Time.

P1 = 24

P2 = 27.

P3 = 31

$= \frac{24+27+31}{3}$

Average Turn around Time = 27.

- **Advantage of FCFS:**
  - The simplest and basic form of CPU scheduling algorithm.
  - Easy to implement, it doesn't require complex data structure.
  - It is well suited for bath systems where the longer time periods for each process are often acceptable.

- **Disadvantage of FCFS:**
  - processes that are at the end of the queue, have to wait longer to Finish.

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- The average waiting time in the FCFS is much higher than in the other.

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• Shortest Job First (SJF) Scheduling.

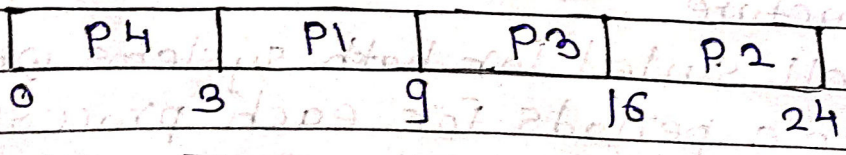
- SJF stands For Shortest Job First.
- In this, the process that has the shortes burst time, execute First.
- SJF is both preemptive and non-preempti

Example.

Process	Burst Time
P1	6
P2	8.
P3	7
P4	3.

Calculate average waiting Time.
Calculate $^{avg}\downarrow$ Turn around Time

→ Gantt chart → is a horizontal bar chart demostrate CPU scheduling



- waiting Time.

P4 = 0

P1 = 3

P3 = 9

P2 = 16

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$$\frac{0+3+9+16}{4} = 7$$

- Turn around Time.

$P_4 = 3$

$P_1 = 9$

$P_3 = 16$

$P_2 = 24$

$$= \frac{3+9+16+24}{4}$$

$$= 13$$

- **Advantages.**
  - It reduces the average waiting time.
  - probably optimal with regard to average turnaround time.
  - SJF Scheduling gives the lowest average waiting time for a specific set of process.

- **Disadvantages.**
  - It cannot be implemented at the level of short-term CPU scheduling.
  - There is no way to know the length of the next CPU burst.

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- **priority Scheduling.**
  - priority scheduling is a non-preemptive algorithm & one of the most common Scheduling algorithms in both system.
  - Each process is assigned a priority. process

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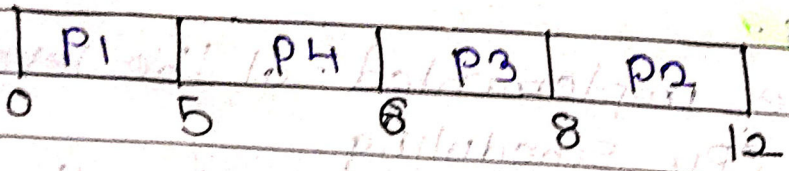
with higher priority is to be executed First & so on.
processes with same priority are executed on first come first served basis
priority can be decided based on memory requirements, time requirements or any other resource requirement

Example. (Higher the no higher priority.

Priority	process	A.T	B.T
1	P1	0	5
2	P2	1	4
3	P3	2	2
4	P4	4	1

Gantt chart - is a horizontal bar chart demostrate CPU Sheduling



P2, P3

Ready Queue

Completion Time

C.T	T.A.T (C.T-A.T)	W.T (T.A.T-B.T)	R.T
5	5-0=5	5-5=0	0 (0)
12	12-1=11	11-4=7	7 (8-1)
8	8-2=6	6-2=4	4 (6-2)
6	6-4=2.	2-1=1.	1

Waiting Time.

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$= A.T\ A.T = \frac{5+11+6+2}{4} = 6$

$A.W.T = \frac{0+7+4+1}{4}$

$= 3.$

Advantages.

priority-based Scheduling ensores that high-priority processes are executed First, which can faster completion of critical tasks.
priority Scheduling can reduce the average waiting time for processes that require a significant amount of time.

Disadvantages.

priority Inversion -
Increase overhead.
Complexity in priority Assignment.

Round Robin Scheduling Algorithm.

Round Robin scheduling algorithm is a method used by operating system to manage the execution time of multiple processes that are competing for CPU attention.
The primary goal of this scheduling method is to ensure that all processes are given an equal opportunity to execute, promoting fairness among task.

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- process Arrival - processes enter the system & are placed in a queue.
- Time Allocation - Each process is given a certain amount of CPU time a quantum.
- Execution - The process use the CPU for the allocated time.

- **Advantages**
  - Fairness
  - simplicity
  - Responsiveness.

- **Disadvantages**
  - overhead
  - underutilization.

Example -

Process	Burst Time
P1	21
P2	3
P3	6
P4	2

- Calculate average waiting Time (Time quantum
- Calculate Average turn around Time.

→ Gantt chart - is a horizontal bar chart demostrat CPU sheduling.

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P1	P2	P3	P4	P1	P3	P1	P1	P1
0	5	8	13	15	20	21	26	31	32

- Arrival Time (C.T - B.T)

P1 = 32 - 0 = 32
P2 = 8 - 0 = 8
P3 = 21 - 0 = 21
P4 = 15 - 0 = 15

$= \frac{32+8+21+15}{4} = \frac{76}{4} = 19$

- Waiting Time (T.A.T - B.T)

P1 = 32 - 21 = 11
P2 = 8 - 3 = 5
P3 = 21 - 6 = 15
P4 = 15 - 0 = 15

$= \frac{11+5+15+13}{4}$

$= \frac{44}{4}$

$= 11$

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	preemptive scheduling	Non-preemptive
1)	preemptive scheduling is more complex.	Simple, but very inefficient.
2)	Example - RR method	Example - FCFS method
3)	It also affects the design of the operating system kernel.	It does not affects the design of OS kernel
4)	It is suitable for Real time system.	It is not suitable for Real time system
5)	It has high overhead	It has low overhead.
6)	Throughput is less.	Throughput is high.
7)	It is cost associated	It does not not cost associated.
8)	Its CPU utilization is very high.	It CPU utilization is very low.
9)	Algorithm design is Complex.	Algorithm design is Simple
10)	It does not treat all processes are equal	It treats all processes equal.